

Insurance Invoice Auditing

Approach and measurement

I built a two-stage pipeline. First, GLM-5.3 reads each hospital's Markdown contract and converts it into one canonical schema containing services, rates, units, effective dates, premiums, discounts, bundles, exclusions, quantity limits, amendments, calculation order, and source evidence. Python validates that extraction before it can be used: schema, quoted evidence, source coverage, service references, numeric values, dates, amendments, and calculation order. Invalid chunks are repaired or the run stops visibly.

Second, the invoice engine maps free-text descriptions to canonical services, selects the effective rules, and calculates expected amounts using integer cents and half-up rounding. It also checks invoice arithmetic, duplicates, bundles, exclusion windows, daily caps, and cumulative thresholds. The LLM interprets language; deterministic Python owns validation, pricing, and enforceable business rules.

Hospital 1 development results

Measure	Result
Labelled invoices evaluated	913
Classification accuracy	100.0%
Precision / recall / F1	100.0% / 100.0% / 100.0%
Exact totals, all invoices	98.7% (901/913)
Exact totals, erroneous invoices	79.3% (46/58)

Hospital 1 was the labelled development set. I compared every prediction with its label at invoice level and measured accuracy, precision, recall, F1, and exact corrected-total accuracy. The classification result found every labelled erroneous invoice without false positives. Reporting 46/58 separately avoids hiding corrected-amount errors behind the 855 correct invoices whose totals are unchanged.

These are development results, not an unbiased estimate for a new hospital. Hospitals 2-5 have no labels, so I report coverage rather than unsupported accuracy: the updated submission contains all **3,942** unseen invoices (1,125 H2; 932 H3; 835 H4; 1,050 H5). Every contract passed the same evidence, numeric, reference, date, and coverage gates before pricing.

Uncertainty and limitations

Contract extraction is the highest-risk step because one confidently wrong rate can affect many invoices. Every financial or enforceable fact therefore cites exact source evidence. A validated extraction is reused only when contract content, provider, model, prompt, and schema versions still match.

The initial leaderboard result exposed a concrete version of this risk: five Hospital 2 rates quoted the correct rows but converted pounds to cents twice. I added deterministic comparison of extracted rates, percentages, thresholds, caps, bundles, exclusion windows, and multipliers against the cited text, plus targeted repair and regression tests. Without using unseen labels, Hospital 2's flag rate fell from 64.5% to 6.8%.

A final error analysis found that weak text matches were also being forced onto unrelated services. Raising the abstention threshold introduced no H1 false positives, improved unknown-service recall from 1/12 to 9/12, and raised exact corrected totals among erroneous invoices from 65.5% to 79.3%.

Remaining limitations

- A description that omits its distinguishing clinical meaning may not be recoverable by deterministic matching.
- One discrepancy can reasonably receive both a generic price mismatch and a specific premium, discount, or unit diagnosis.
- Malformed dates prevent reliable amendment or weekend-rule selection; these cases are flagged rather than assigned an invented rate.
- Duplicate allocation and cumulative discounts require stable chronological ordering where a contract does not specify tie-breaking.
- Hospital 1's perfect classification is not proof of cross-contract generalisation; H2-H5 need independent labels for that claim.

What I would do differently with another week

- Benchmark smaller, cheaper models and add an explicit deterministic availability fallback that emits the same schema, passes the same validation gates, records its provenance, and carries lower confidence.
- Build an adjudicated set of ambiguous descriptions and measure matcher top-1 accuracy, abstention, and confidence calibration; invoke an LLM resolver only for the low-margin queue.
- Lock a Hospital 1 holdout and add contract-perturbation tests so prompt and rule-engine changes are not measured only on examples used during development.
- Obtain a small independently labelled sample from each unseen hospital, add more contract-shape regression tests, and refine category priority while retaining detailed contributing findings in the audit trace.